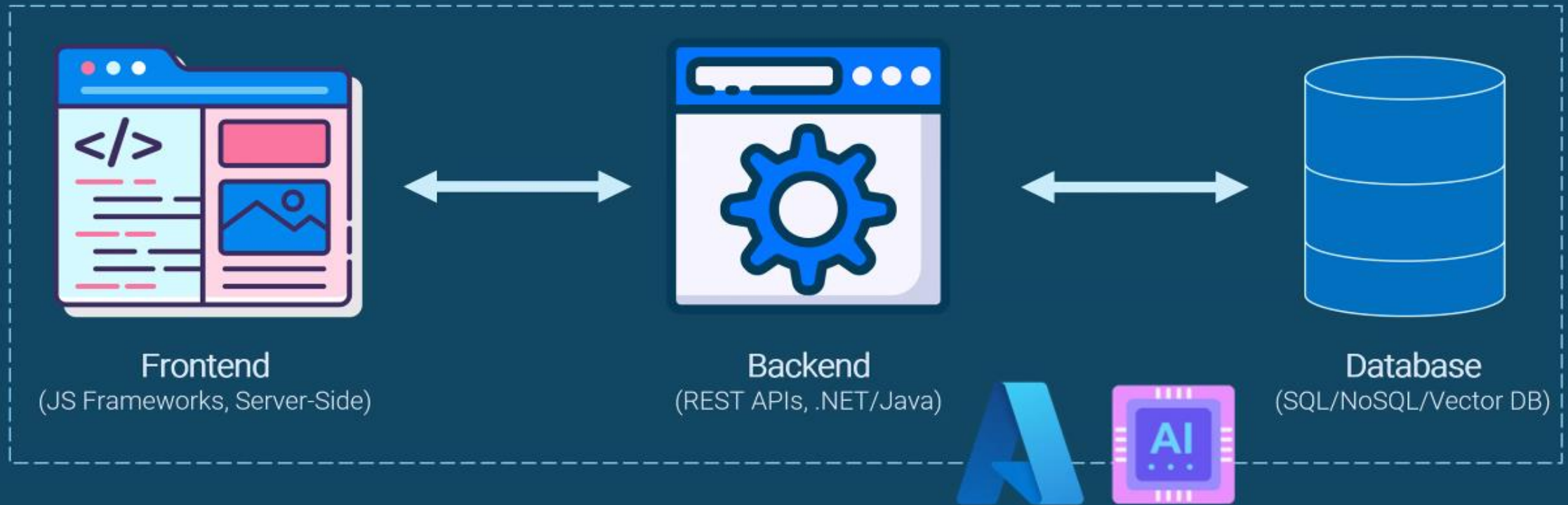
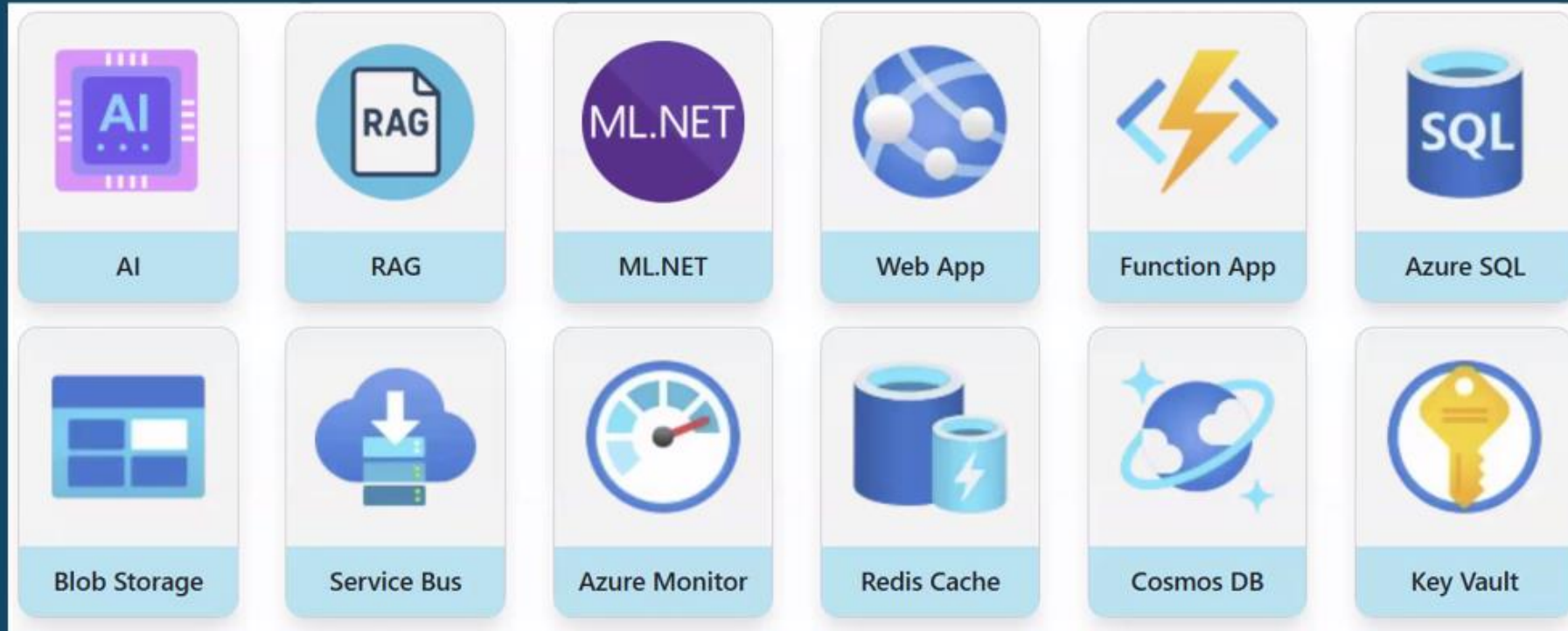


What is Full-Stack AI Development?

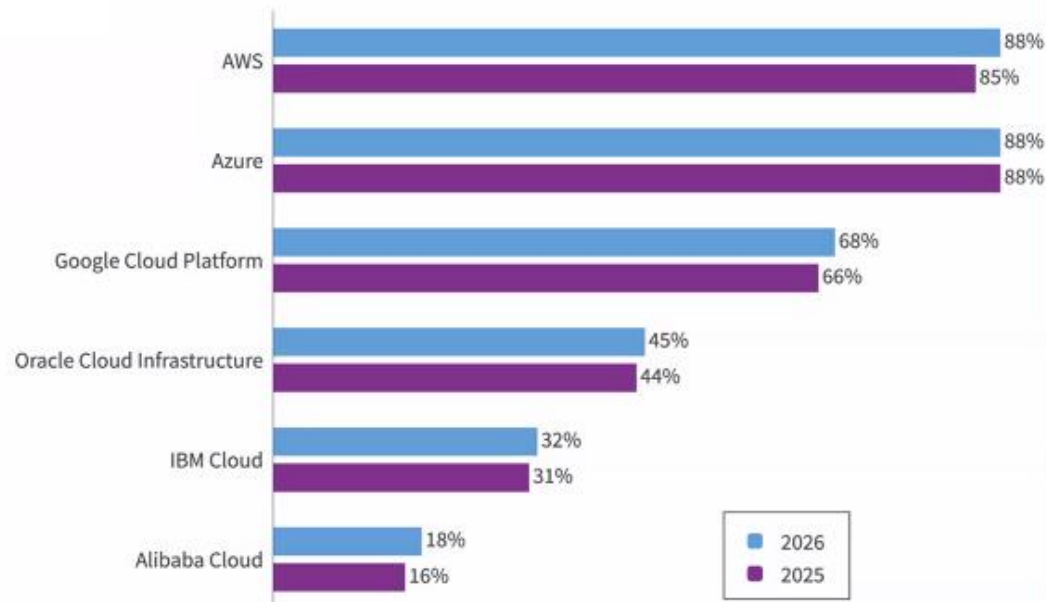


Microsoft Azure and AI/ML



Cloud Usage Report: 2026-2025

YoY public cloud provider usage by all organizations

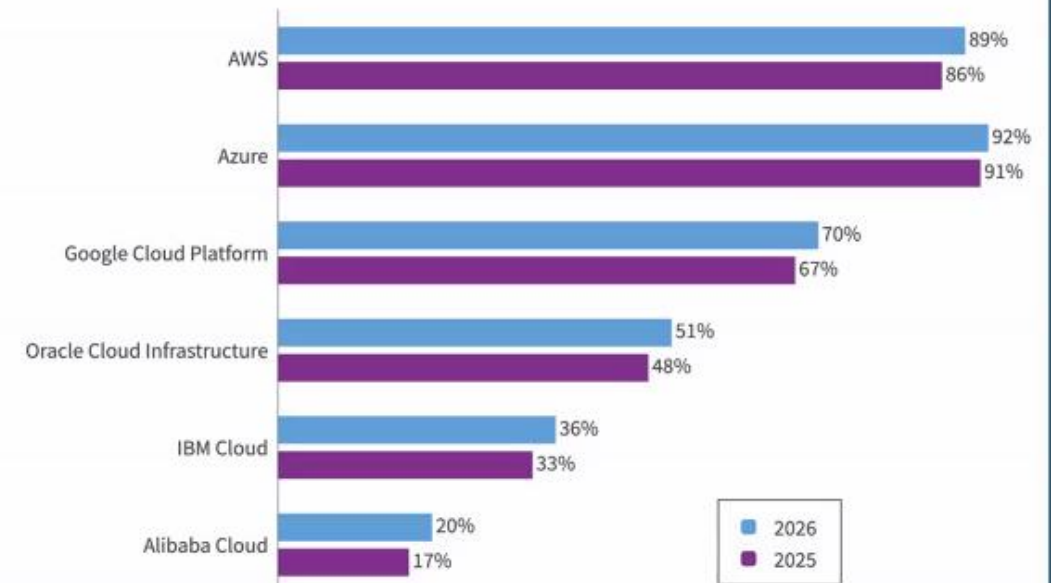


2026: N=753, 2025: N=759

Source: Flexera 2026 State of the Cloud Report (Figure 32)

flexera

YoY enterprise use of public cloud providers



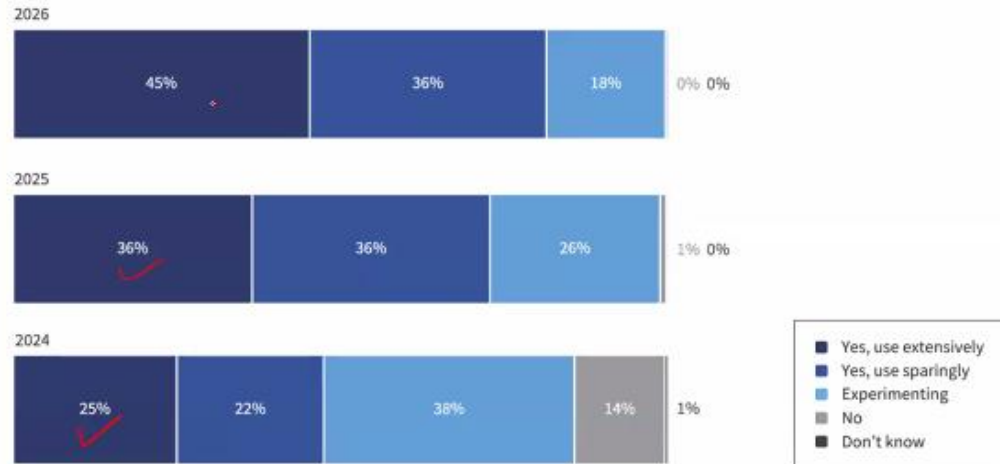
2026: N=620, 2025: N=622

Source: Flexera 2026 State of the Cloud Report (Figure 34)

flexera

Cloud AI Services Usages: 2026-2025-2024

Use of generative AI (GenAI) public cloud services



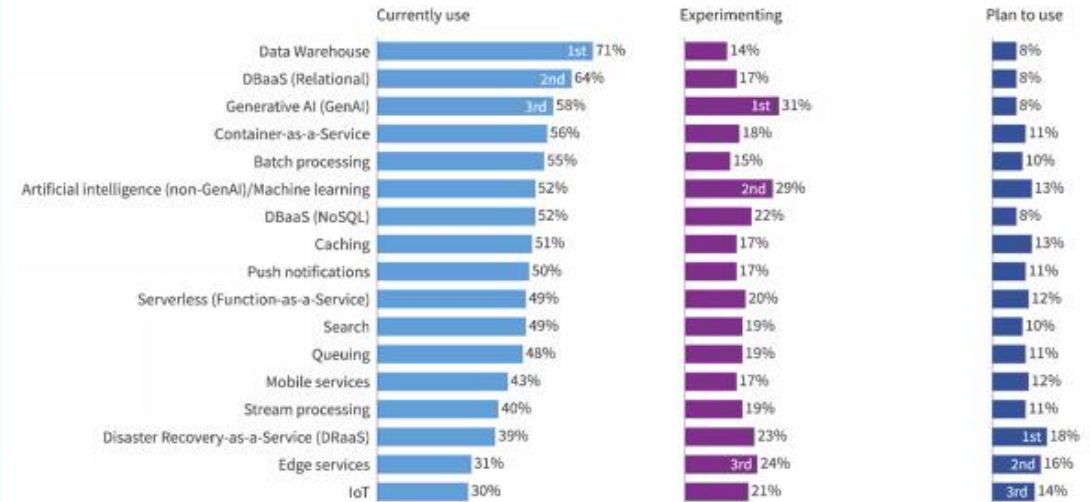
2026: N=692, 2025: N=628; 2024: N=753

Source: Flexera 2026 State of the Cloud Report (Figure 42)

flexera



Public cloud services used by all organizations



N=753

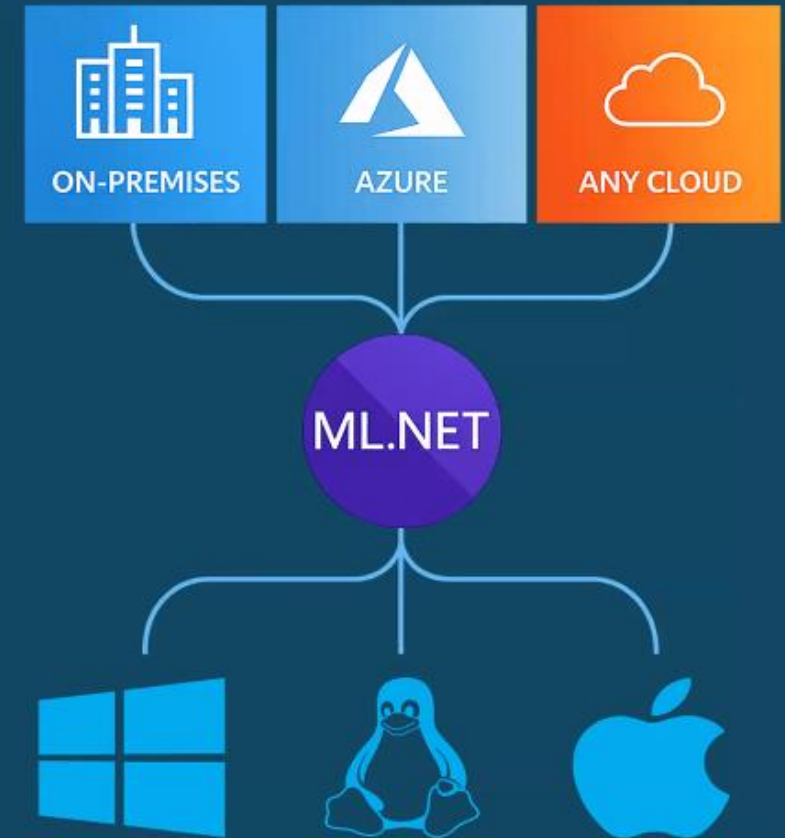
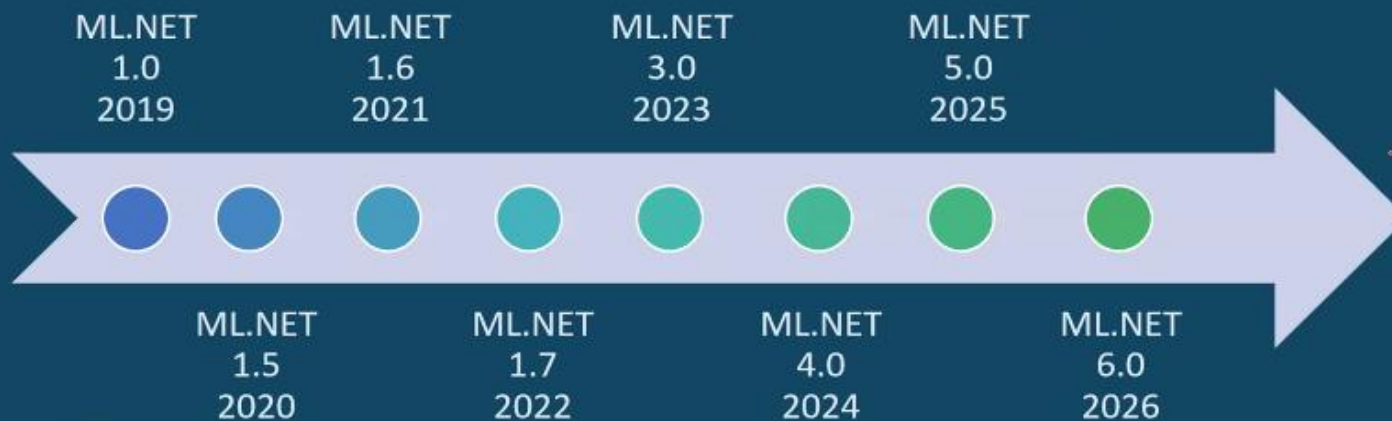
Source: Flexera 2026 State of the Cloud Report (Figure 41)

flexera



What is ML.NET?

- An open-source, cross-platform ML framework
- Designed specifically for .NET developers
- Supports C# & F# for Machine learning
- Runs on Linux, Windows, macOS & any Cloud
- Go from zero ML knowledge to production-ready models in < 30 minutes



Why ML.NET?

- Most enterprise data & logic already lives in .NET world
- Avoid context switching (Python ↔ C# microservices pain)
- Reuse .NET skills: Integrate ML into web, mobile, desktop, games, and IoT apps
- No need to learn Python or R
- Seamless integration with existing .NET libraries and tools
- High performance and accuracy for large datasets
- Strong typing, debugging, familiar tools (Visual Studio/Rider/VS Code)

.NET AI Extensions

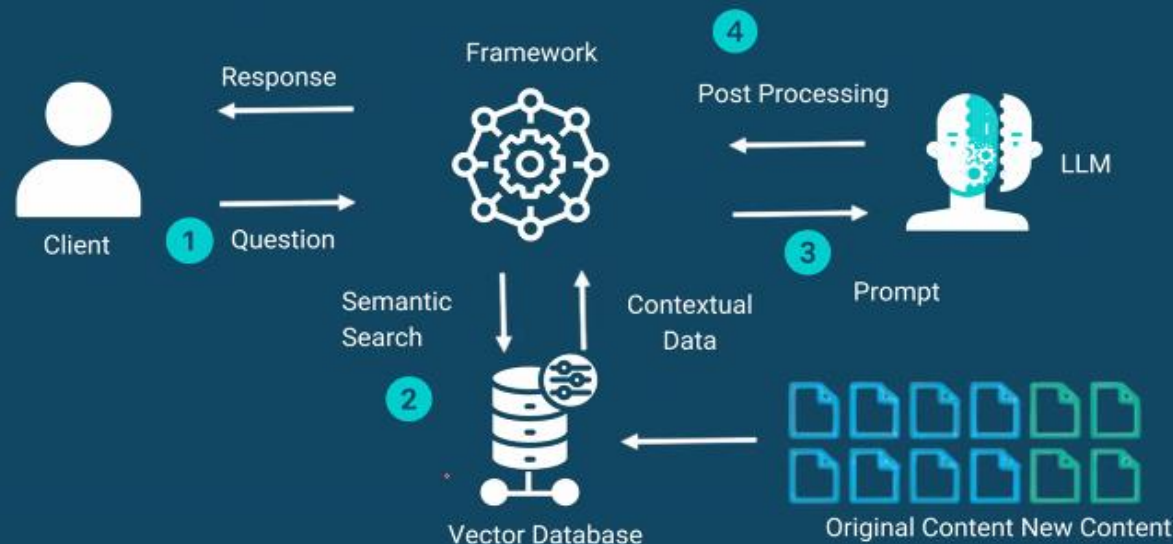
- Modern AI libraries for integrating Generative AI, LLMs, embeddings, and AI services into .NET applications.
- Works with Popular Gen AI, LLMs and embeddings:
 - Azure OpenAI
 - OpenAI
 - Ollama
 - Semantic Kernel
 - Vector databases

Challenges Enterprises Face Today

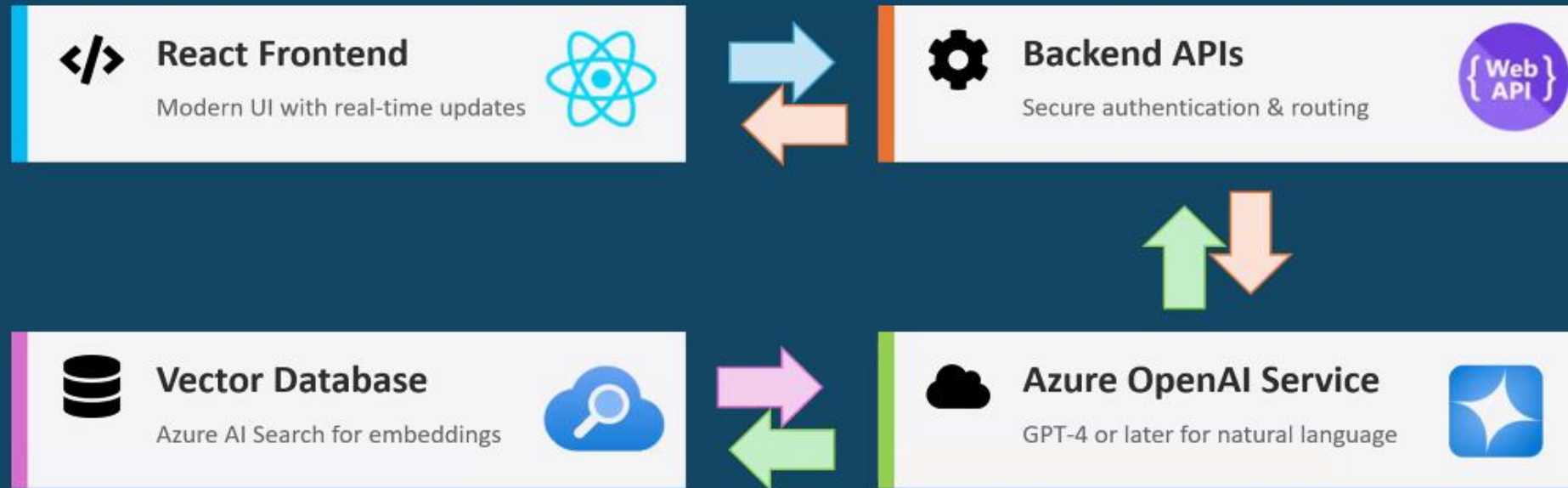
-  Repetitive questions – Agents answer the same FAQs 60% of the time, wasting hours daily.
-  Scattered knowledge – Information is in emails, PDFs, CRMs, databases, and wikis with no central place to find it.
-  Slow responses – Customers wait hours or days for basic answers, leading to frustration and anger.
-  Inconsistent answers – The same question gets different replies across chat, email, and phone, breaking customer trust.
-  No smart scaling – More customers mean hiring more agents.

Solution: RAG = Retrieval + Context + Generation

- An AI pattern that bring your data into the LLM at runtime to generate accurate, context-aware response.
- 🔍 **Retrieval System** : Semantic search over knowledge base (vector DB)
- 🧠 **Context Injection** : Add retrieved data into prompt
- 🤖 **Generation Model (LLM)** : Produces grounded, accurate answer

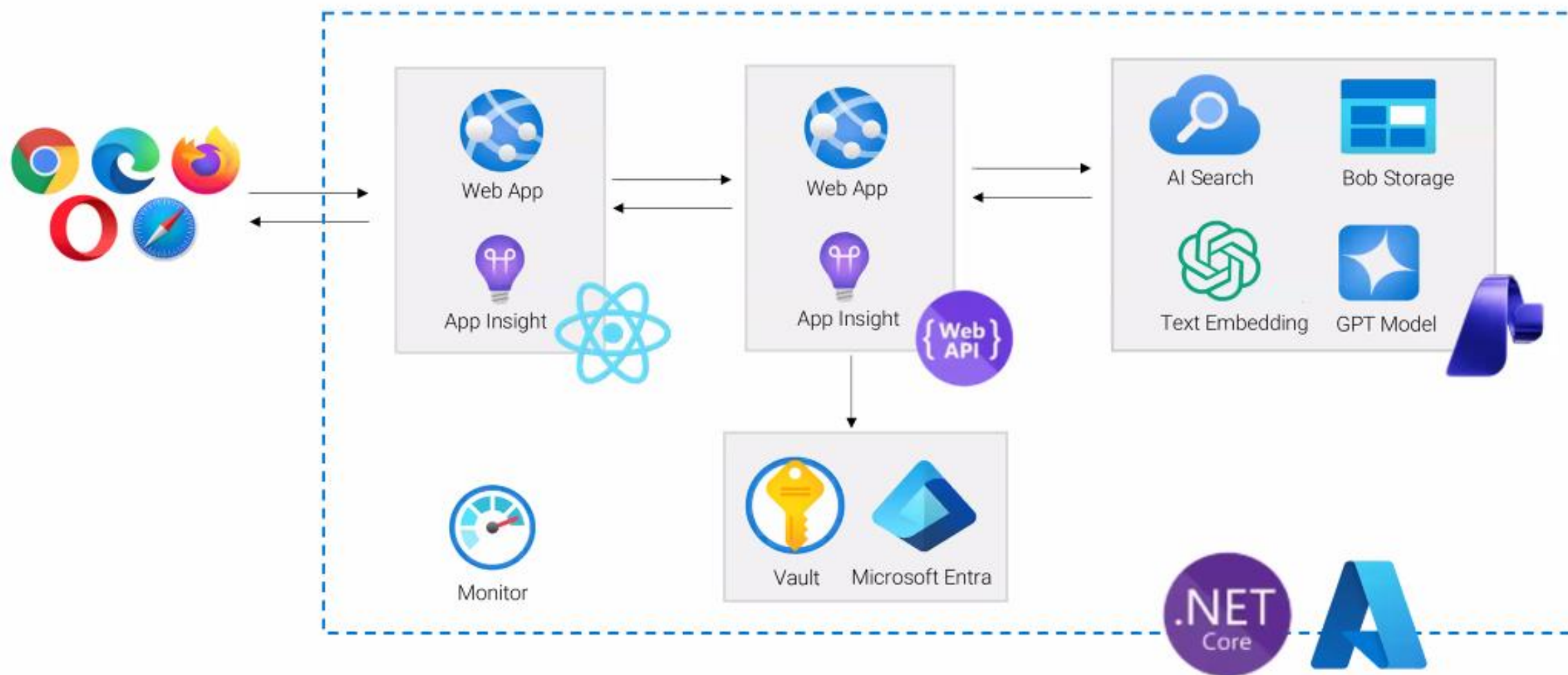


ScholarHat AI Chat App Architecture



Data Flow: User Query → React UI → Backend API → Azure OpenAI + Vector Search → Intelligent Response

Azure CHAT GPT Infrastructure



scholaropenaix - Microsoft Azure

Models + endpoints - Azure OpenAI

Ask Gemini

ai.azure.com/resource/deployments?wsid=/subscriptions/4cc400bc-8706-4682-b513-3a847ab32c7f/resourceGroups/scholarai-rg/providers/Microsoft.Cog...

Search Star Profile Menu

Microsoft Foundry | Azure OpenAI

scholaropenaix

Models + endpoints

New Foundry

All resources

scholaropenaix (eastus, S0)

Help

Model catalog

Playgrounds

Chat

Assistants PREVIEW

Video PREVIEW

Audio PREVIEW

Images

Tools

Fine-tuning

Azure OpenAI PREVIEW

Evaluation

Stored completions PREVIEW

Batch jobs

Monitoring

Shared resources

Deployments

Quota

Guardrails + Controls

Risks + alerts PREVIEW

Data files

Assistant vector stores PREVIEW

Model deployments

Model deployments App

+ Deploy model

Table with 2 columns: Name, Model retirement date

Deploy gpt-5-mini

Deployment name * gpt-5-mini

Deployment type Global Standard

Global Standard: Pay per API call with the highest rate limits. Learn more about Global deployment types.

Data might be processed globally, outside of the resource's Azure geography, but data storage remains in the AI resource's Azure geography. Learn more about data residency.

Deployment details

Model version 2025-08-07

Capacity 100K tokens per minute (TPM)

Content safety DefaultV2

AI resource scholaropenaix

Resource location East US

Version upgrade policy Once a new default version is available

Customize

Deploy Cancel

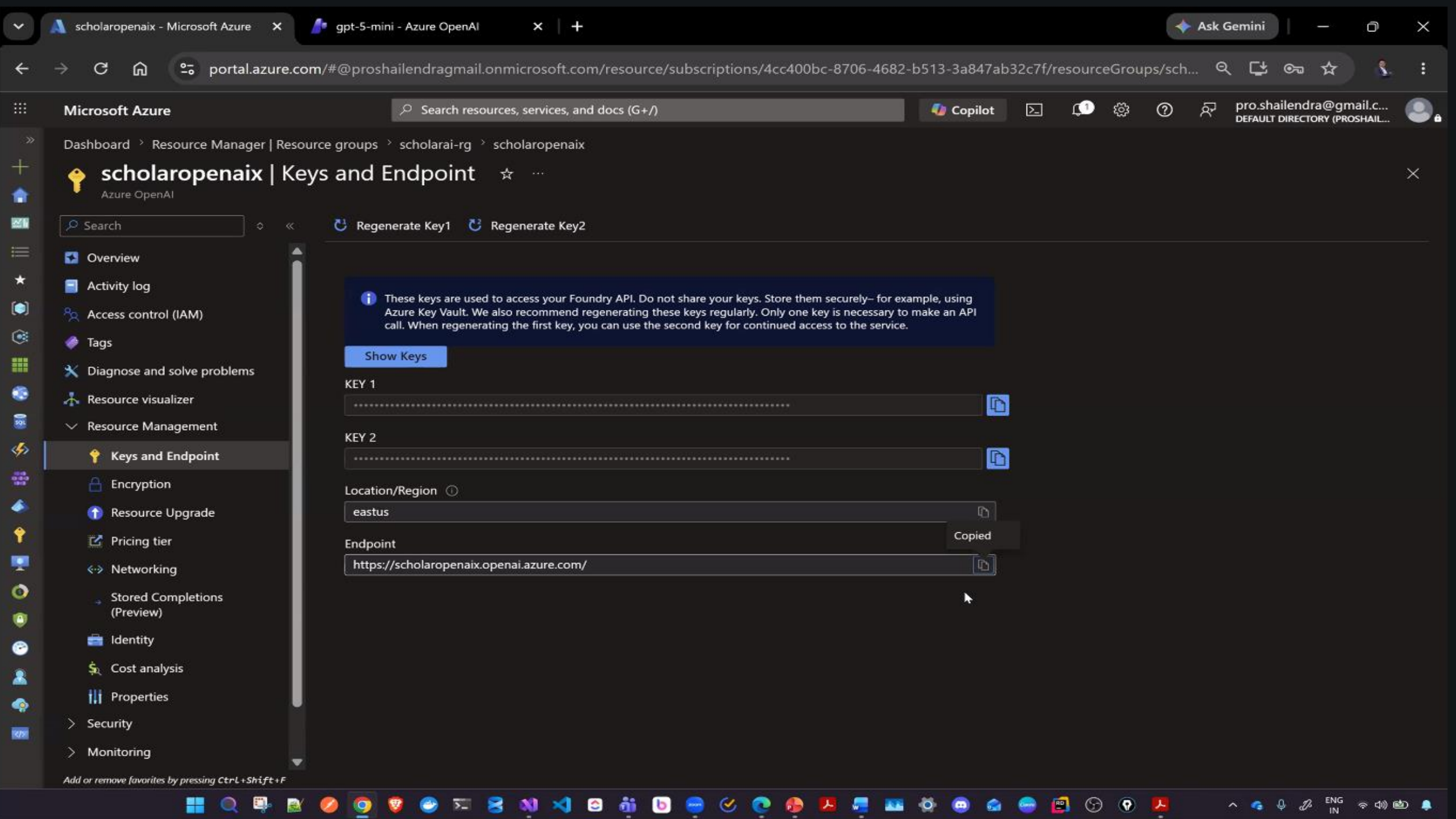
Columns

Model retirement date

Content filter

Taskbar with various application icons

System tray with ENG IN, network, and other status icons



FileEditViewGitProjectBuildDebugTestToolsExtensionsWindowHelpSearchwebapi

ChatResponse.csAzureOpenAISettings.csProgram.csChatHistoryService.cswebapi.httpAzureOpenAI...ntroller.csChatRequest.csappsettings.json

Schema: obj\Debug\net10.0\appsettingsschema.json

```
1 {
2   "Logging": {
3     "LogLevel": {
4       "Default": "Information",
5       "Microsoft.AspNetCore": "Warning"
6     }
7   },
8   "AllowedHosts": "*",
9   "AzureOpenAI": {
10    "Endpoint": "https://scholaropenaix.openai.azure.com/",
11    "ApiKey": "5q3dFG7MYrpR0JVfVmgHCok94QCLsv2MdxGyVsRnLywccqWQPCrtJQQJ99CGACYeBjFXJ3w3AAABACOGMCow",
12    "DeploymentName": "gpt-5-mini"
13  }
14 }
15
```

94 % 1 0 12, Ch: 30 SPC CRLF UTF-8

Error ListOutputPackage Manager ConsoleDeveloper PowerShellGitHub Copilot TerminalDeveloper PowerShellGitHub Copilot Terminal

Solution Explorer

Search Solution Explorer (Ctrl+)

Solution 'webapi' (1 of 1 project)

- Webapi
 - Connected Services
 - Dependencies
 - Properties
 - Configuration
 - AzureOpenAISettings.cs
 - Controllers
 - AzureOpenAIController.cs
 - Models
 - ChatRequest.cs
 - ChatResponse.cs
 - Services
 - ChatHistoryService.cs
 - appsettings.json
 - Program.cs
 - webapi.http

Find and Replace

Properties

Git Changes

FileEditViewGitProjectBuildDebugTestToolsExtensionsWindowHelpSearch

webapi

DebugAny CPUContinue

Application Insights

GitHub Copilot Chat

ChatResponse.csAzureOpenAISettings.csProgram.csChatHistoryService.cswebapi.httpAzureOpenAI...ntroller.csChatRequest.csappsettings.json

POST {{webapi_HostAddress}}/azure/globalenv: <none>

1 @webapi_HostAddress = https://localhost:7136/api

2

3 Send request | Debug

4 POST {{webapi_HostAddress}}/azure/global

5 Accept: application/json

6 Content-Type: application/json

7 {

8 "question":"What is Azure?",

9 "conversationId":"1"

10 }###

11

200 OKTime: 11.37 sSize: 2.2 KB

BodyHeadersRaw

{

"answer": "Microsoft Azure (commonly called "Azure") is Microsoft

"conversationId": "1"

}

application/json; charset=utf-8

Solution Explorer

Search Solution Explorer (Ctrl+;)

Solution 'webapi' (1 of 1 project)

External Sources

Webapi

Connected Services

Dependencies

Properties

Configuration

AzureOpenAISettings.cs

Controllers

AzureOpenAIController.cs

Models

ChatRequest.cs

ChatResponse.cs

Services

ChatHistoryService.cs

appsettings.json

Program.cs

webapi.http

94 %No issues found

Ln: 11, Ch: 1SPCCRLFUTF-8

Solution E...Git ChangesPropertiesNotificatio...

AutosLocalsWatch 1Developer Command PromptDeveloper PowerShellCall StackBreakpointsException SettingsCommand WindowImmediate WindowOutputGitHub Copilot TerminalContainersGitHub Copilot TerminalGitHub Copilot Terminal

Request completed...

0/042mainazure-ai

Windows taskbar with various application icons and system tray icons.

FileEditSelectionViewGoRun

chatService.ts

src > services > chatService.ts > sendChatMessage > response

```
1  s from "axios";
2  { Mode } from "../types";
3
4  _URL = "https://scholarhatai-api.azurewebsites.net/api";
5  RL = window.location.host === 'localhost:3000' ? 'https://localhost:7136/api' : 'https://scholargpt-api-gpb6hag9fbh7eha
6
7  t sendChatMessage = async (question: string, mode: Mode) => {
8    == "custom") {
9      esponse = await axios.post(`${API_URL}/azure/custom`, {
10      ion,
11
12
13      response.data;
14      (mode == "image") {
15        esponse = await axios.post(`${API_URL}/azure/image`, {
16        ion,
17
18
19      response.data;
20
21      esponse = await axios.post(`${API_URL}/azure/global`, {
22      ion,
23
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

AZURE

QUERY RESULTS

Deploy to app "scholargpt" Succeeded in 10s

Zip and deploy workspace "d:\GitHub\azure-ai\projects\scholarhat-gpt-dotnet\frontend\dist" 6s

Build app "scholargpt" in Azure 4s

AZURE

RESOURCES Remote

Azure Subscription

App Services

edTechSparkAPI QuotaExceeded

evideoprimeapi QuotaExceeded

scholargpt Deploying...

Application Settings

Code Optimizations

Deployment Slots

Deployments

Files

Logs Read-only

WebJobs

scholargpt-api

scholarhatadmin QuotaExceeded

scholarhatdev QuotaExceeded

Azure Arc-enabled machines

Azure Cosmos DB

Azure Cosmos DB for MongoDB (RU)

Azure DocumentDB

Container Apps

Durable Task Scheduler

Function App

Logic App

Managed Identities

Microsoft Foundry

PostgreSQL servers (Flexible)

PostgreSQL servers (Standard)

Spring Apps

WORKSPACE

ACCOUNTS & TENANTS

HELP AND FEEDBACK

Ln 11, Col 12

Spaces: 2

UTF-8

CRLF

{ } TypeScript

ScholarHat AI

+ New chat

No history yet

Light mode



How can I help you today?
Ask me anything — code, questions, analysis, and more.

tell me about the

Global GPT

shift+enter

SchlarHat GPT - Powered by Azure AI

3. .NET Backend Engineering

ASP.NET Core

LINQ

Entity Framework Core

ADO.NET

WebAPI Core

Guest

Unit Testing

Project #1 - ePizzaHub Food Delivery Platform

4. Frontend Development with React

TypeScript

React - TSX

Vite Tool

React Router

React Hook Form

Yup

Axios

Redux

Unit Testing

Project #2 - OTT Streaming Platform

5. Frontend Development with Angular

Angular CLI

Angular Foundations

RxJS

